

GALELA GAMAR MALAMU, Indonesia

WMO: 974060

Lat: 1.8384N

Lon: 127.7880E

Elev: 52

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	69.5	70.7	67.2	100.3	76.9	68.6	105.2	76.3	15.2	84.0	13.3	83.8	0.7	N/A	0.272

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	14.0	91.3	79.4	90.4	79.3	89.7	79.3	81.9	88.6	81.4	88.1	80.8	87.5	7.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.1	156.3	86.6	79.4	153.0	85.8	78.9	150.4	85.3	45.9	89.0	45.2	88.4	44.6	87.8	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	11.9	10.0	66.7	93.0	2.5	1.2	64.9	93.9	63.4	94.6	62.0	95.2	60.1	96.1		
(5)			65.2	84.1	3.1	1.6	62.9	85.3	61.1	86.2	59.3	87.2	57.1	88.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.4	80.0	79.5	79.9	80.7	81.2	80.4	80.0	80.0	80.3	80.7	80.8	80.9
	DBStd	1.49	1.55	1.47	1.36	1.43	1.20	1.50	1.52	1.52	1.26	1.30	1.38	1.39
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11085	930	826	928	921	967	913	929	930	910	951	923	957
	CDD65	5610	465	406	463	471	502	463	464	465	460	486	473	492
	CDH74	50448	4039	3423	4105	4419	4830	4080	3983	3990	4133	4529	4366	4551
CDH80	17681	1312	1076	1424	1524	1728	1395	1358	1406	1543	1722	1576	1617	
Wind	WSAvg	2.7	2.6	2.5	2.6	2.7	2.5	2.3	2.5	2.5	2.9	3.1	2.9	2.8
Precipitation	PrecAvg	87.3	7.8	7.1	7.7	8.6	8.9	10.3	6.7	6.4	4.8	5.6	7.2	6.6
	PrecMax	141.4	17.2	21.2	34.3	20.5	15.6	18.9	16.5	15.1	10.2	10.4	14.4	19.2
	PrecMin	40.1	1.5	0.1	0.9	2.4	2.4	3.9	0.0	0.0	0.0	0.0	0.0	1.3
	PrecStd	25.2	4.3	5.1	7.2	4.4	3.5	4.3	4.1	4.3	3.4	3.3	3.7	4.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.2	89.9	90.1	91.2	91.3	90.6	91.1	90.9	91.6	92.4	92.2	91.6
		MCWB	80.4	79.5	78.9	80.1	80.2	80.1	79.0	79.5	77.7	76.8	80.9	81.2
	2%	DB	89.0	88.5	88.9	89.7	90.0	89.3	89.5	89.6	90.2	90.7	90.0	89.9
		MCWB	79.8	78.9	78.4	79.8	79.8	79.6	78.9	78.8	78.1	78.1	79.8	80.3
	5%	DB	87.8	87.2	87.9	88.5	88.8	88.2	88.1	88.2	89.0	89.4	88.8	88.7
		MCWB	79.3	78.4	78.2	79.3	79.5	79.0	78.5	78.3	78.2	78.4	79.2	79.8
	10%	DB	86.4	86.0	86.7	87.2	87.6	86.9	86.7	86.8	87.6	87.8	87.4	87.4
		MCWB	78.7	77.9	77.9	78.9	79.2	78.7	78.2	78.1	78.1	78.3	78.8	79.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.1	81.7	80.9	81.7	82.3	81.8	81.7	81.5	82.0	81.9	82.4	82.6
		MCDB	88.3	87.2	87.8	88.2	88.5	88.3	88.8	88.1	88.2	88.7	90.0	89.9
	2%	WB	81.0	80.4	80.0	80.8	81.2	80.8	80.3	80.2	80.7	80.7	81.3	81.6
		MCDB	87.2	86.4	86.6	87.6	87.4	87.4	87.1	87.2	87.3	87.5	88.3	88.5
	5%	WB	80.1	79.6	79.2	80.0	80.5	80.0	79.4	79.5	79.7	79.9	80.2	80.7
		MCDB	86.1	85.4	85.7	86.5	86.7	86.2	85.9	86.2	86.3	86.4	86.7	87.4
	10%	WB	79.3	78.7	78.4	79.4	79.8	79.2	78.6	78.7	78.9	79.0	79.3	79.8
		MCDB	84.7	84.2	84.6	85.5	85.7	85.0	84.8	85.0	85.2	85.4	85.5	85.9
Mean Daily Temperature Range	5% DB	MDBR	13.5	13.5	14.5	13.8	14.0	14.0	14.4	15.0	15.9	15.7	14.4	13.8
		MCDBR	15.5	15.7	16.2	15.8	15.7	15.8	16.4	16.8	18.2	18.0	16.4	15.9
		MCWBR	7.1	7.3	7.0	6.8	7.0	7.0	7.1	7.5	8.0	7.4	7.1	7.2
	5% WB	MCDBR	14.4	14.3	14.9	14.8	14.6	14.9	15.0	15.5	15.7	15.8	15.7	15.3
		MCWBR	7.6	7.6	7.1	6.9	6.9	7.2	7.1	7.6	7.9	7.6	7.7	7.8
Clear-Sky Solar Irradiance	taub	0.422	0.425	0.422	0.424	0.419	0.416	0.418	0.429	0.440	0.480	0.439	0.425	
	taud	2.485	2.464	2.477	2.466	2.481	2.500	2.496	2.451	2.408	2.276	2.408	2.468	
	Ebn at Noon	287	288	286	280	274	272	272	275	277	268	280	285	
	Edh at Noon	36	37	37	36	35	33	34	36	39	44	38	36	
All-Sky Solar Radiation	RadAvg	1585	1699	1782	1805	1711	1555	1620	1741	1832	1829	1699	1611	
	RadStd	95	161	173	155	103	100	174	157	155	141	120	135	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page